PHYSIOLOGY OF NERVOUS SYSTEM

Autonomic Nervous System



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Nicolas Varela

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- « La vraie générosité envers le futur consiste à tout donner au présent».
- « Real generosity towards the future lies in giving all to the present».

Albert Camus, The Rebel, 1951

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Summary

- Physiological anatomy of Autonomic Nervous System
  - Sympathetic
  - Parasympathetic
- Neurotransmitters and receptors
  - Sympathetic
  - Parasympathetic
- Autonomic Nervous System actions
- Sympathetic and Parasympathetic tone

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Anatomy of S-ANS

- Sympathetic chain
- Prevertebral ganglia
  - Celiac
  - Sup/inf mesenteric
  - Aortico-renal
  - Hypogastric
- Preganglionic: ACh
- Postganglionic: Nad (except ACh for sweat glands)
- Splanchnic nerve
- Spinal nerve (piloerector…)

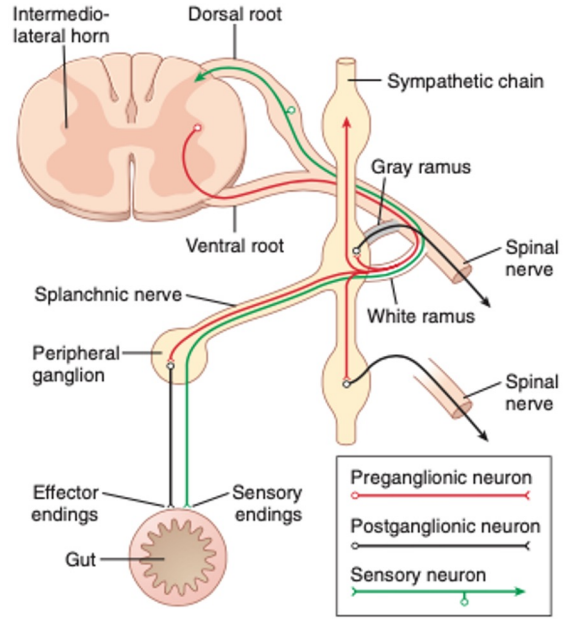


Figure 61-2. Nerve connections among the spinal cord, spinal nerves, sympathetic chain, and peripheral sympathetic nerves.

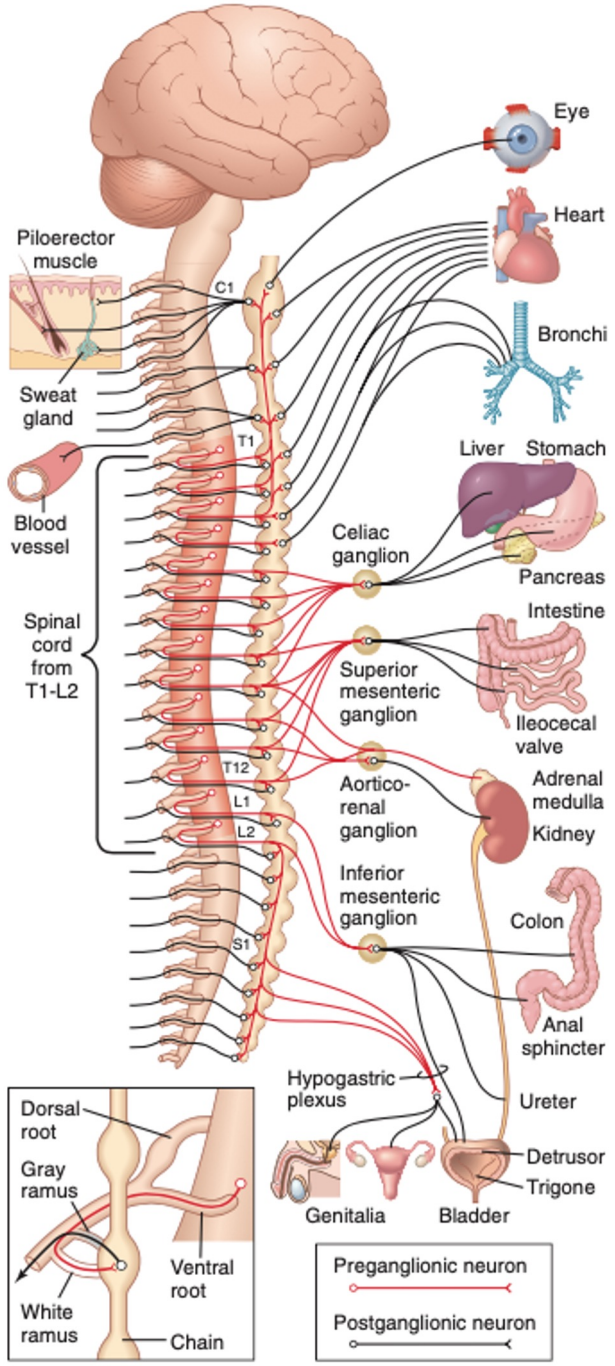


Figure 61-1. Sympathetic nervous system. The black lines represent postganglionic fibers, and the red lines show preganglionic fibers.

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Anatomy of P-ANS

- Cranial nerves III, VII, IX, X
- Sacral nerves
- 75% P-ANS: vagus nerve
- Preganglionic/postganglionic: ACh

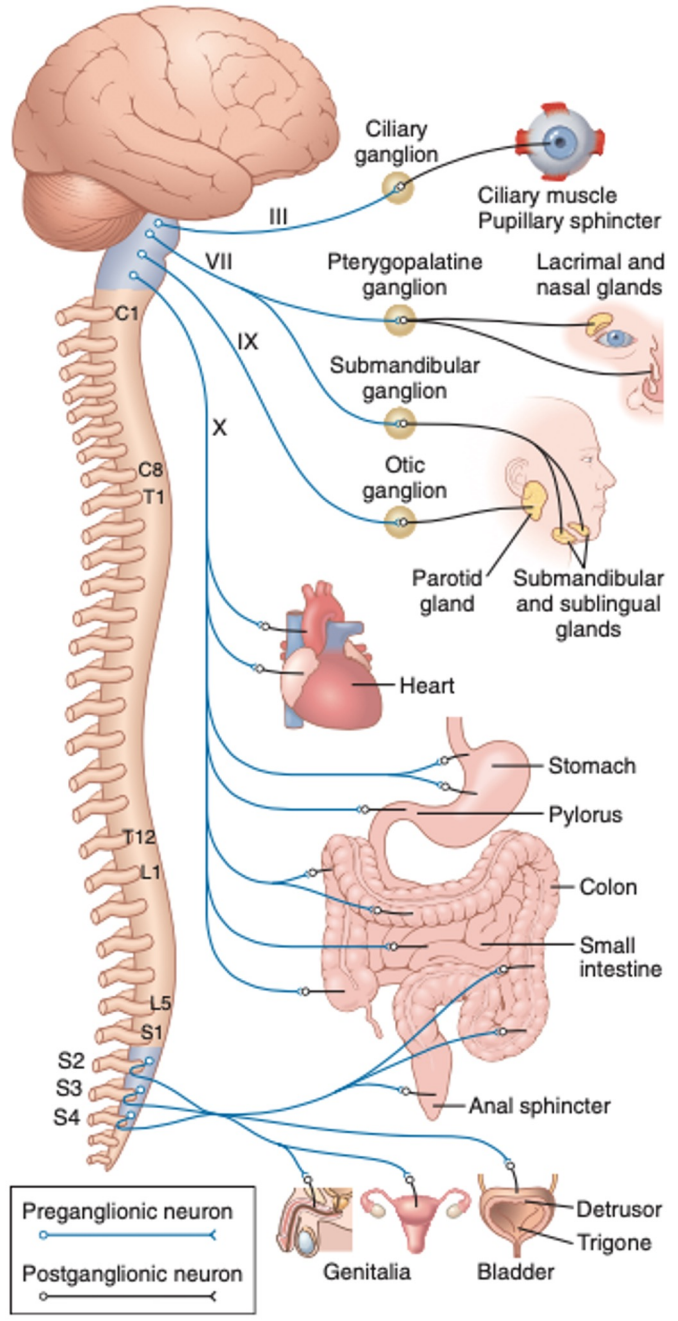


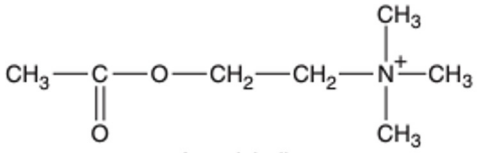
Figure 61-3. The parasympathetic nervous system. The blue lines represent preganglionic fibers and the black lines show postganglionic fibers.

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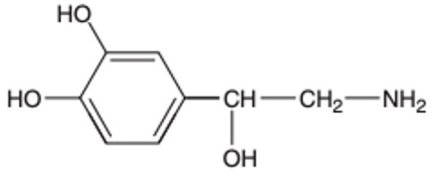
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Acetylcholine and Norepinephrine



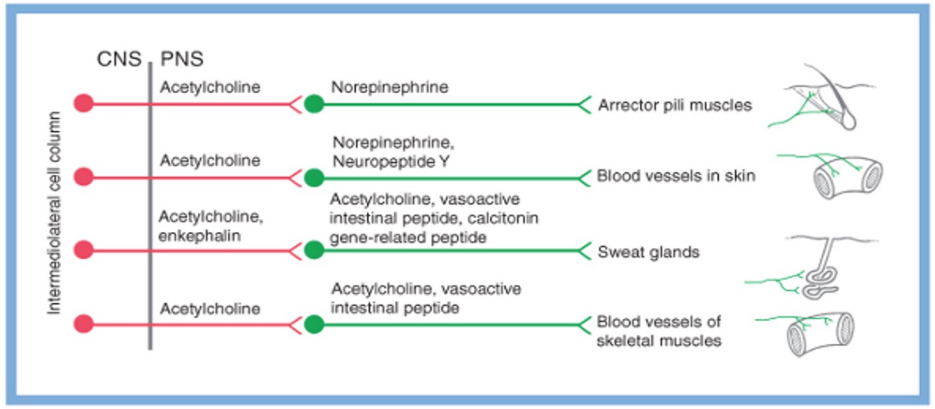
Acetylcholine



Norepinephrine

- ACh: cholinergic fibres / parasympathetic transmitter
- NAd (i.e NE): adrenergic fibres / sympathetic transmitter

- ALL PREGANGLIONIC: cholinergic



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Symp



PS



- POSTGANGLIONIC:
  - S-ANS: mainly adrenergic (except sweat glands and some blood vessels)
  - P-ANS: cholinergic

- Adrenal gland

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Cholinergic receptors



- Muscarinic:
  - Activated by Muscarine (poison from toadstools)
  - G-Protein mechanism
  - Postganglionic cholinergic P and S
  - M1-M5



Clitocybe dealbata

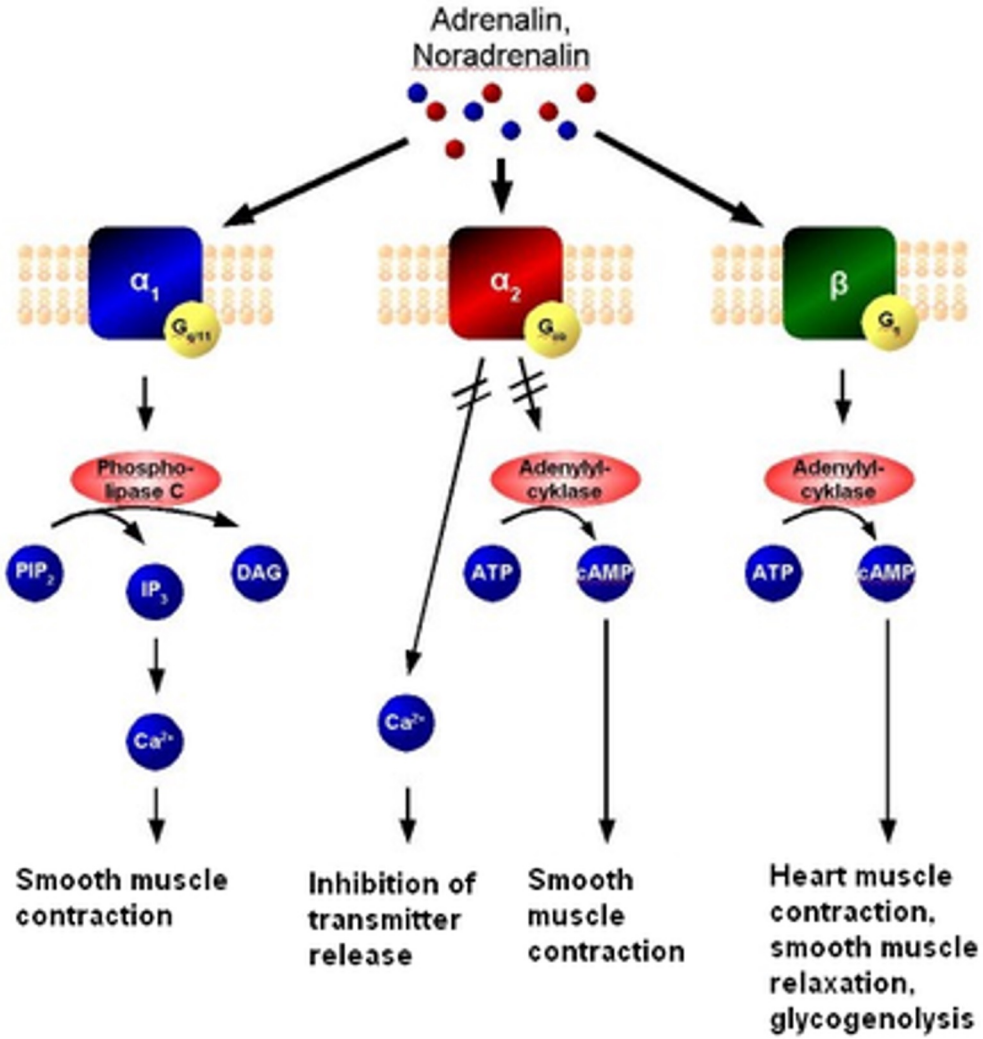
- Nicotinic:
  - Activated by Nicotine
  - Ligand-gated ion channels
  - Autonomic ganglia (pre-post junction) P and S
  - N1: skeletal muscle / N2: ganglia



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Adrenergic receptors MOA

- Alpha and Beta
  - Alpha: 1 and 2
  - Beta: 1, 2 and 3
- G protein coupled
- Activated by NAd and Ad
  - NAd: Mainly Alpha
  - Ad: Alpha and Beta ($\beta 2 > \beta 1 > \alpha$)



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Adrenergic receptors location

- Alpha 1:
  - Vascular smooth muscles
  - GI Tract and Bladder
- Alpha 2:
  - Presynaptic
  - Negative feedback (inhibition)
- Beta 1:
  - Heart
  - Salivary glands/fat/kidney
- Beta 2:
  - Bronchial/bladder/GI smooth muscle
- Beta 3:
  - Fat

Alpha Receptor	Beta Receptor
Vasoconstriction	Vasodilation (β_2)
Iris dilation	Cardioacceleration (β_1)
Intestinal relaxation	Increased myocardial strength (β_1)
Intestinal sphincter contraction	Intestinal relaxation (β_2) Uterus relaxation (β_2)
Pilomotor contraction	Bronchodilation (β_2)
Bladder sphincter contraction	Calorigenesis (β_2)
Inhibits neuro-transmitter release (α_2)	Glycogenolysis (β_2) Lipolysis (β_1) Bladder wall relaxation (β_2) Thermogenesis (β_3)

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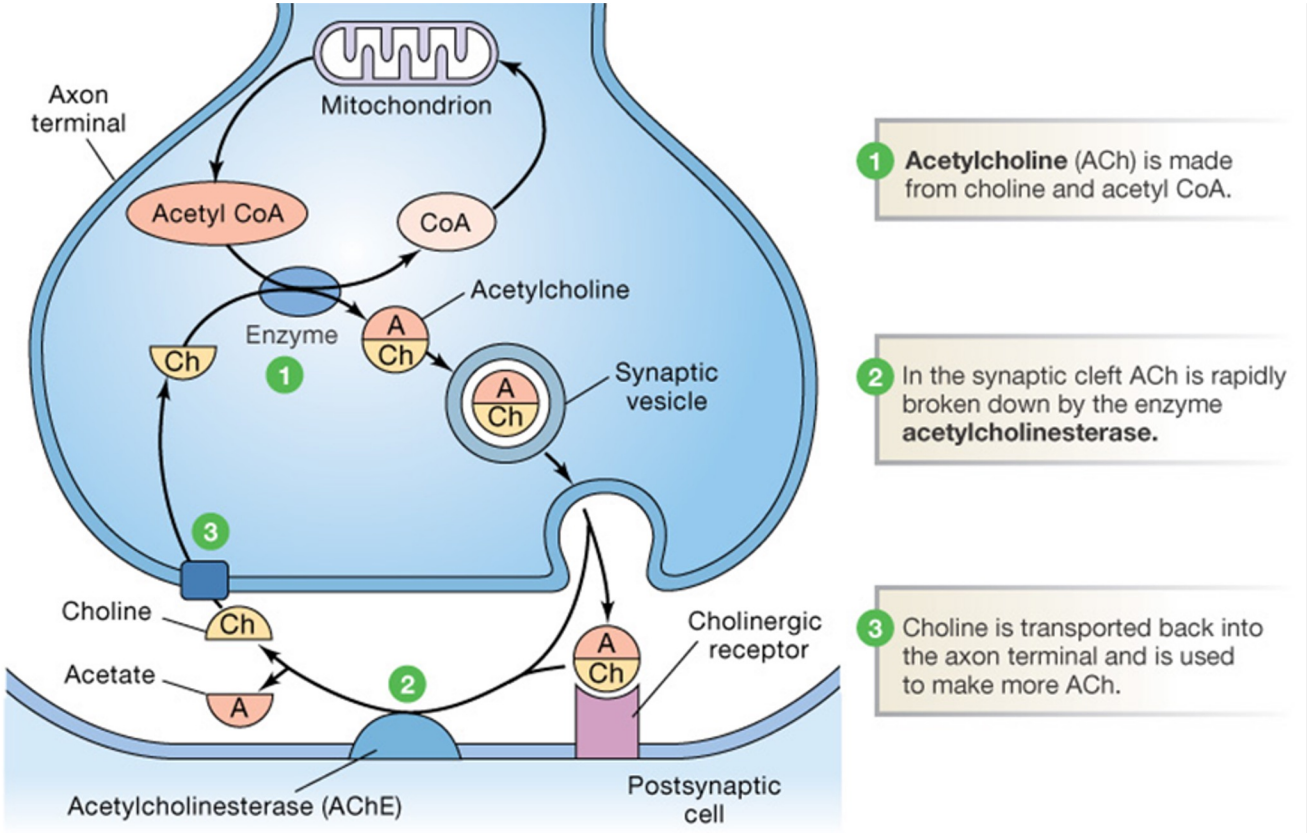
Pseudoefedrine and Phenylefrine



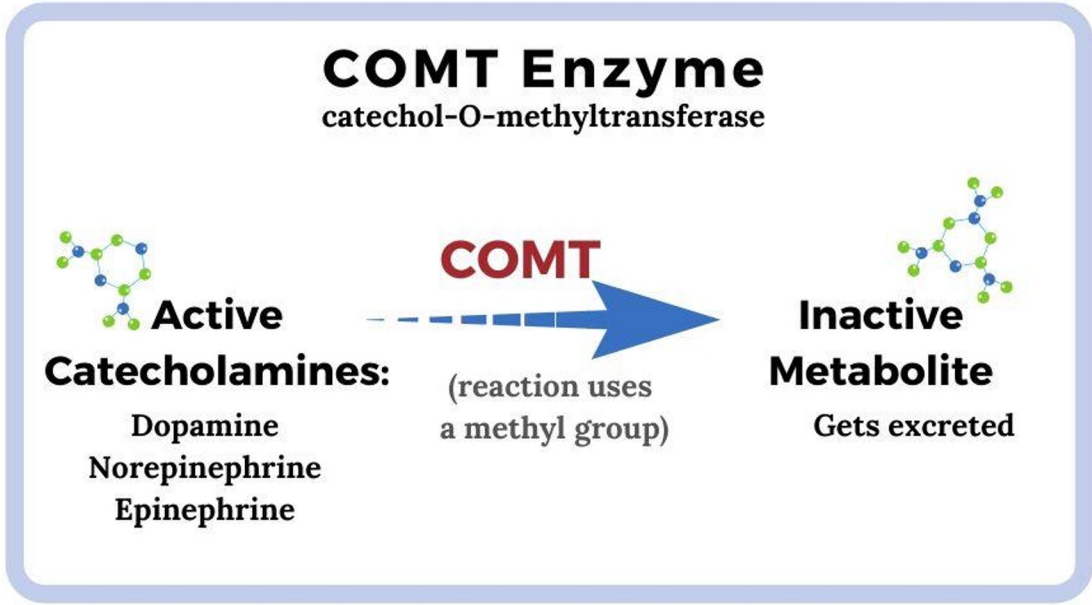


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Neurotransmitters elimination

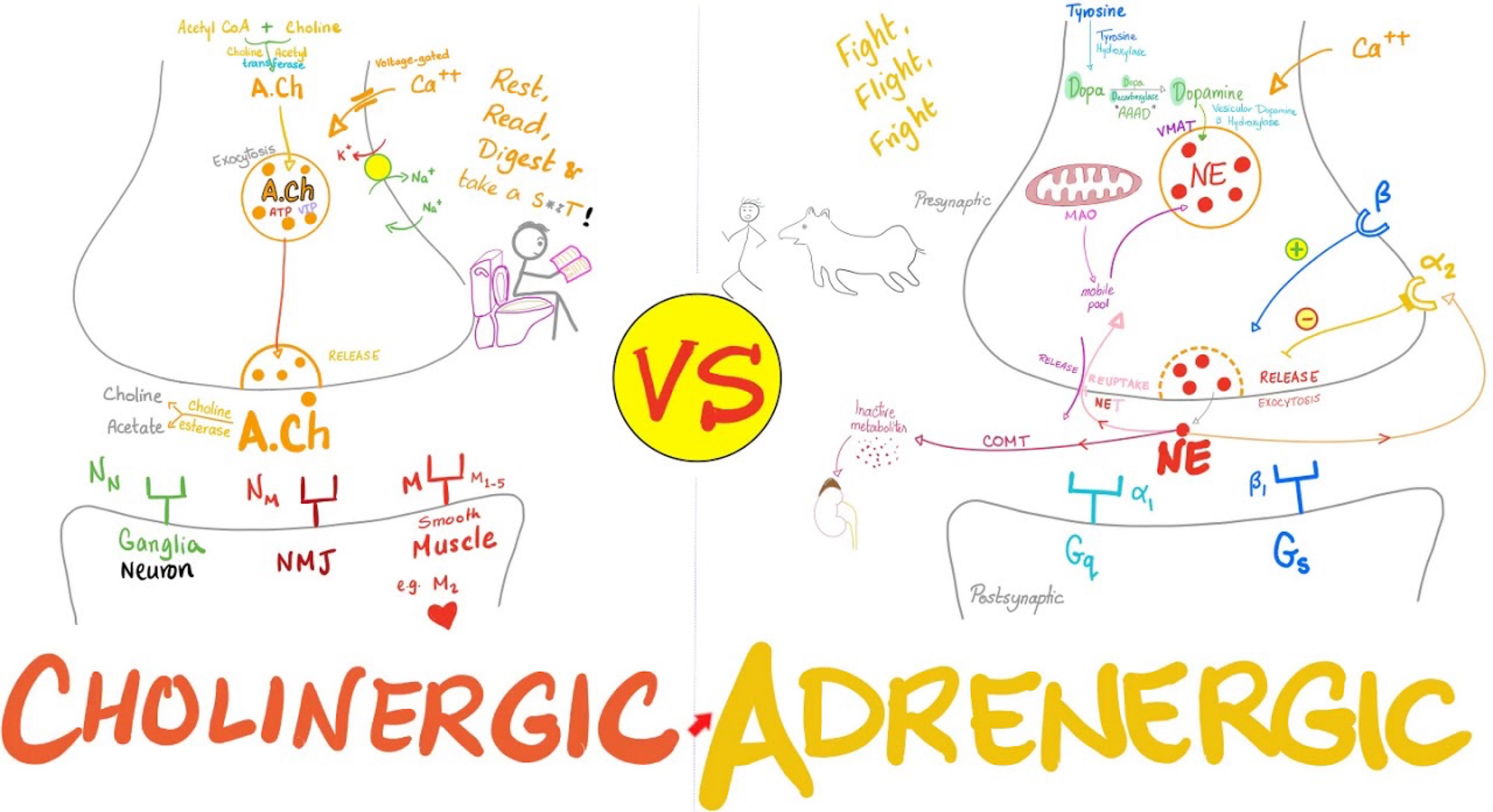


1. **Acetylcholine** (ACh) is made from choline and acetyl CoA.
2. In the synaptic cleft ACh is rapidly broken down by the enzyme **acetylcholinesterase.**
3. Choline is transported back into the axon terminal and is used to make more ACh.



ACh

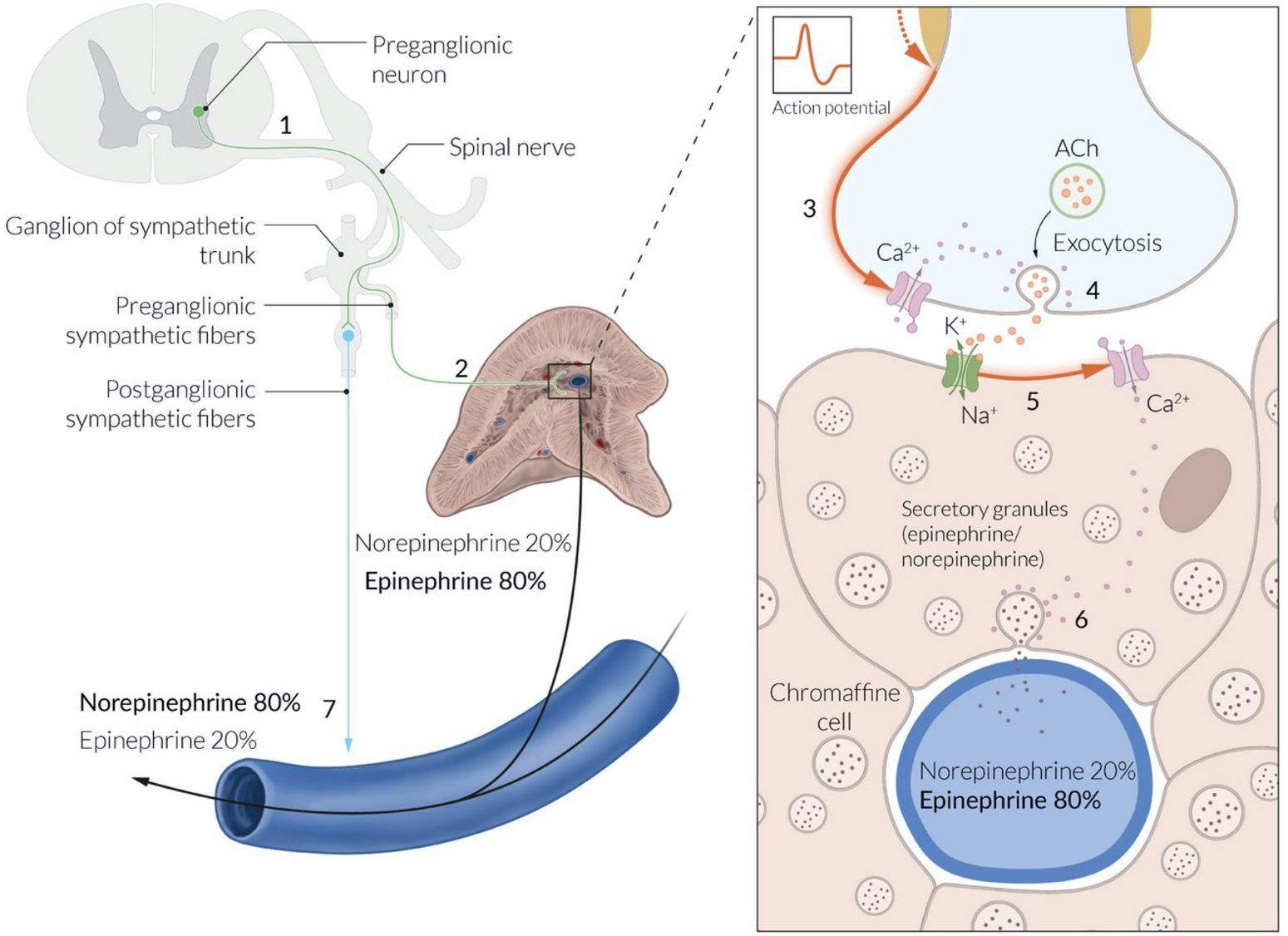
NAd


Acetyl CoA + Choline
Choline acetyl transferase
A.Ch
Exocytosis
A.Ch
ATP VIP
Voltage-gated
Ca++
K+
Na+
Na+
Rest, Read, Digest & take a S*zT!
RELEASE
Choline
Acetate
Choline esterase
A.Ch
NN
NM
M
M1-5
Ganglia
Neuron
NMJ
Smooth Muscle
e.g. M2
VS
Fight, Flight, Fright
Tyrosine
Tyrosine Hydroxylase
Dopa
Dopa Decarboxylase
"AAAD"
Dopamine
Vesicular Dopamine β Hydroxylase
VMAT
Ca++
NE
MAO
Presynaptic
β
α2
+
−
mobile pool
RELEASE
REUPTAKE
NET
RELEASE
EXOCYTOSIS
Inactive metabolites
COMT
NE
α1
β1
Gq
Gs
Postsynaptic
CHOLINERGIC
ADRENERGIC

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Adrenal medulla

- Preganglionic fibres enter adrenal medulla
- Direct excretion to blood stream of catecholamines



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ANS Actions

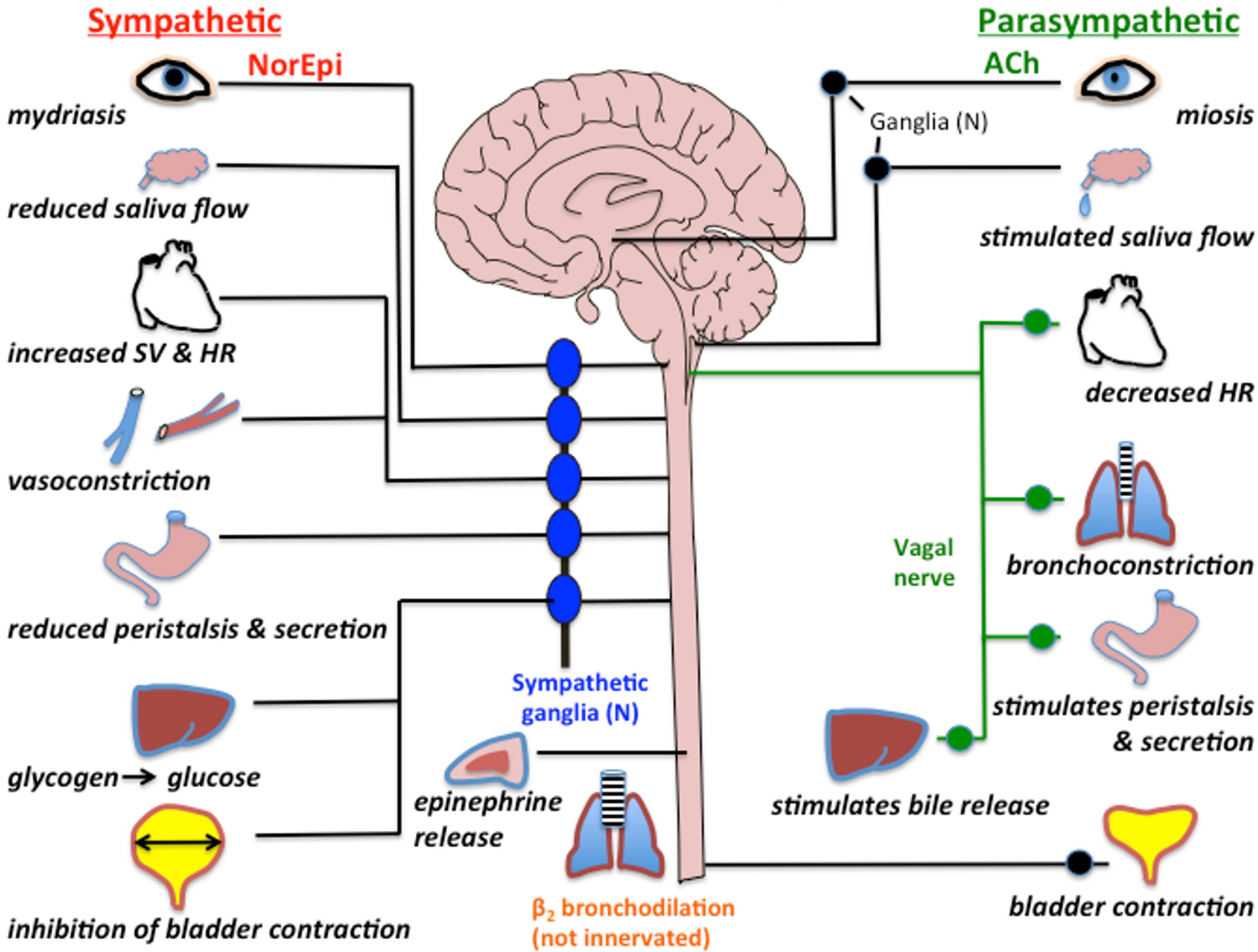
- Symp and Parasymp are usually opposite

- Distribution may be unequal
  - Smooth muscle / Kidney: Symp
  - Genitalia: both but differently located
  - Heart: even distribution

- Sympathetic: strength, energy, movement, “fight or flight”
- Parasympathetic: repair, restore, relax, “rest and digest”

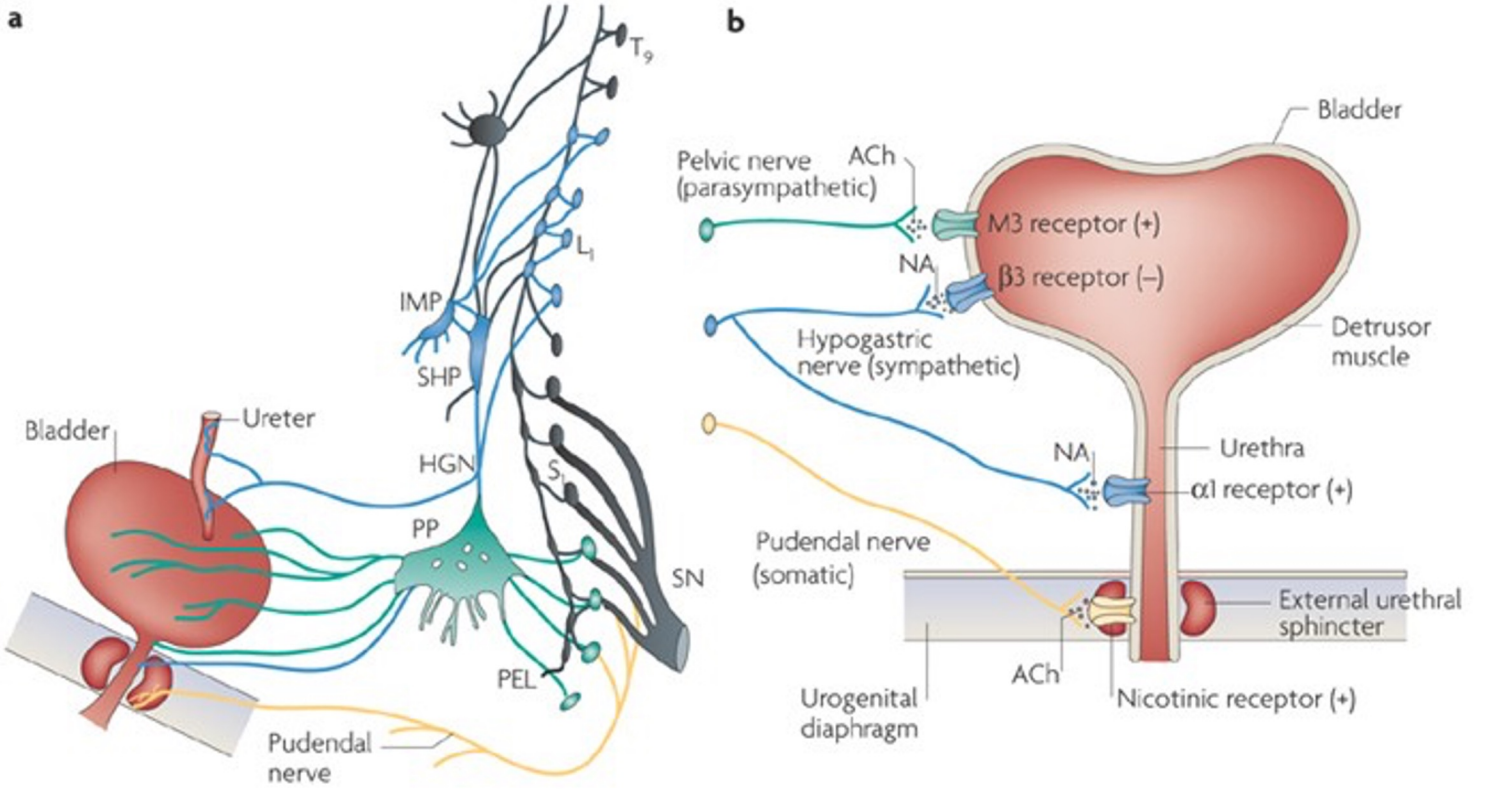
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The Autonomic Nervous System



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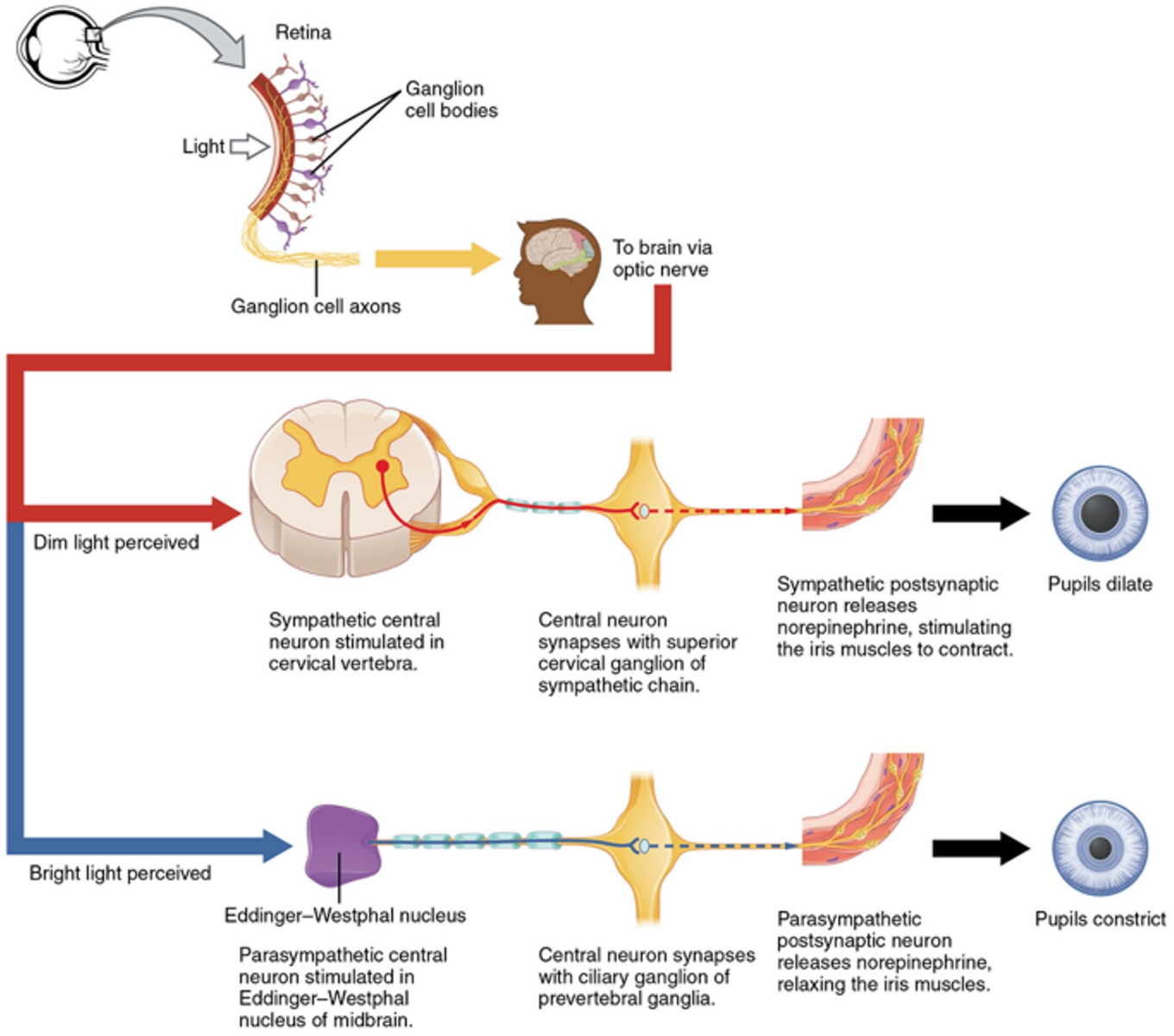
Neural control of micturition



Nature Reviews | Neuroscience

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Pupil contraction



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Actions

SYMPATHETIC NERVOUS SYSTEM

↑ HEART RATE

↑ BLOOD PRESSURE

↓ DIGESTION

"FIGHT OR FLIGHT"

PARASYMPATHETIC NERVOUS SYSTEM

↓ HEART RATE

↑ DIGESTION

"REST & DIGEST"

	Sympathetic		Parasympathetic	
Organ	Action	Receptor	Action	Receptor
Heart				
SA node, heart rate	↑	β_1	↓	M
AV nodal conduction	↑	β_1	↓	M
Contractility	↑	β_1	↓ (atria only)	M
Vascular Smooth Muscle				
Skin; splanchnic	Constricts	α_1		
Skeletal muscle	Dilates	β_2		
Skeletal muscle	Constricts	α_1		
Endothelium			Releases EDRF	M
Bronchioles	Dilates	β_2	Constricts	M
Gastrointestinal Tract				
Smooth muscle, walls	Relaxes	α_2, β_2	Contracts	M
Smooth muscle, sphincters	Contracts	α_1	Relaxes	M
Saliva secretion	↑	β_1	↑	M
Gastric acid secretion			↑	M
Pancreatic secretion			↑	M
Bladder				
Wall, detrusor muscle	Relaxes	β_2	Contracts	M
Sphincter	Contracts	α_1	Relaxes	M
Male Genitalia	Ejaculation	α	Erection	M
Eye				
Radial muscle, iris	Dilates pupil (mydriasis)	α_1		
Circular sphincter muscle, iris			Constricts pupil (miosis)	M
Ciliary muscle	Dilates (far vision)	β	Contracts (near vision)	M
Skin				
Sweat glands, thermoregulatory	↑	M*		
Sweat glands, stress	↑	α		
Pilomotor muscle (goose bumps)	Contracts	α		
Lacrimal Glands			Secretion	M
Liver	Gluconeogenesis; glycogenolysis	α, β_2		
Adipose Tissue	Lipolysis	β_1		
Kidney	Renin secretion	β_1		

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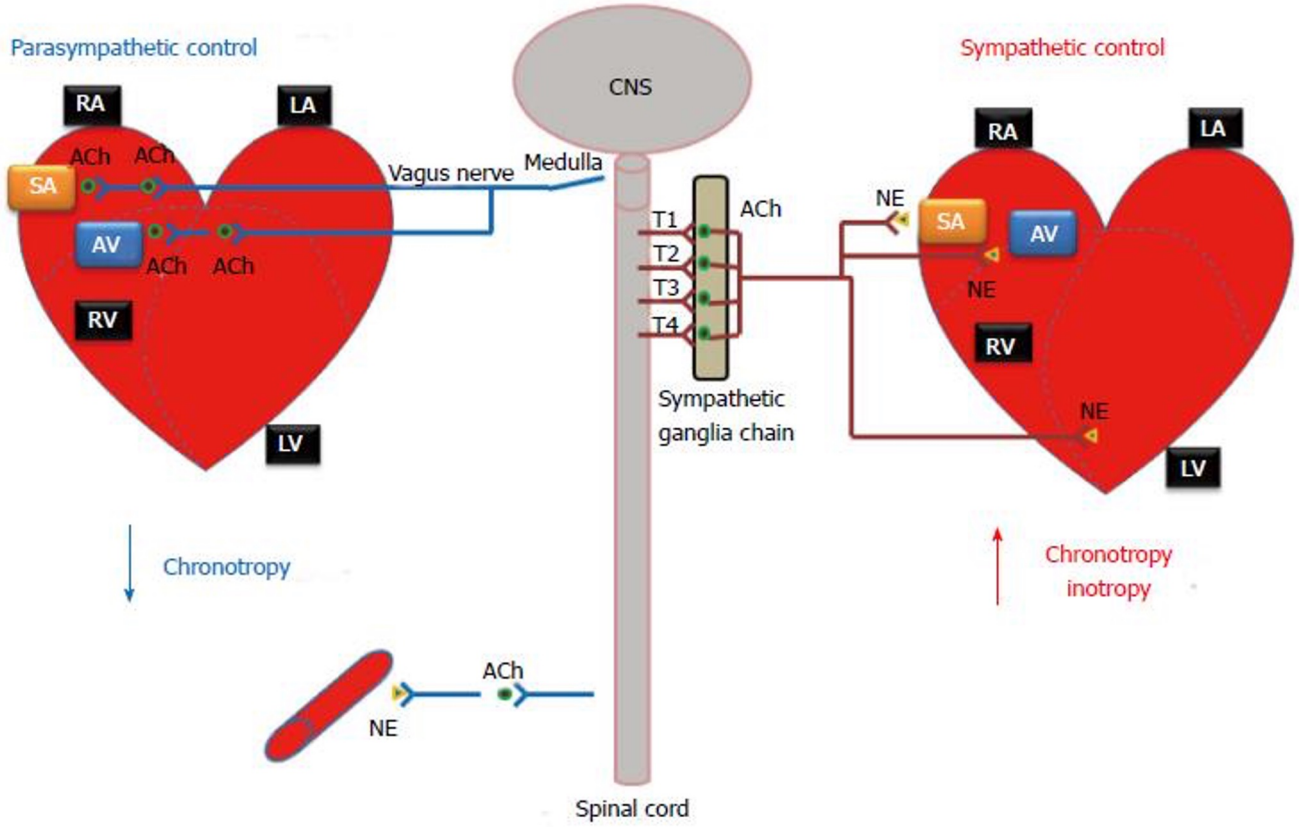
Autonomic Nervous System Tone

- Some organs have uneven autonomic effect
- This is called ANS Tone
- Organs can have more sympathetic or parasympathetic tone

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Heart

- Higher Parasympathetic tone
- P-ANS tends to slow down heart
- Automatism: 100m bpm
- Denervated heart!



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Vascular tone

- Only Sympathetic
- Normally contracted (circulating cathecolamines + S-ANS)
- Can increase contraction

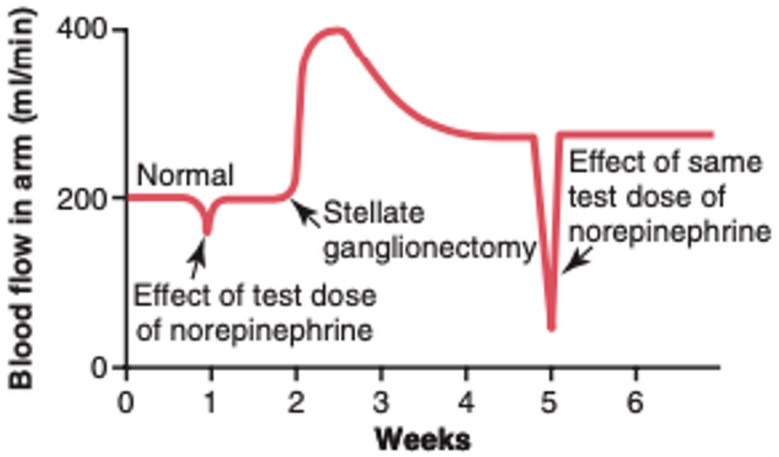
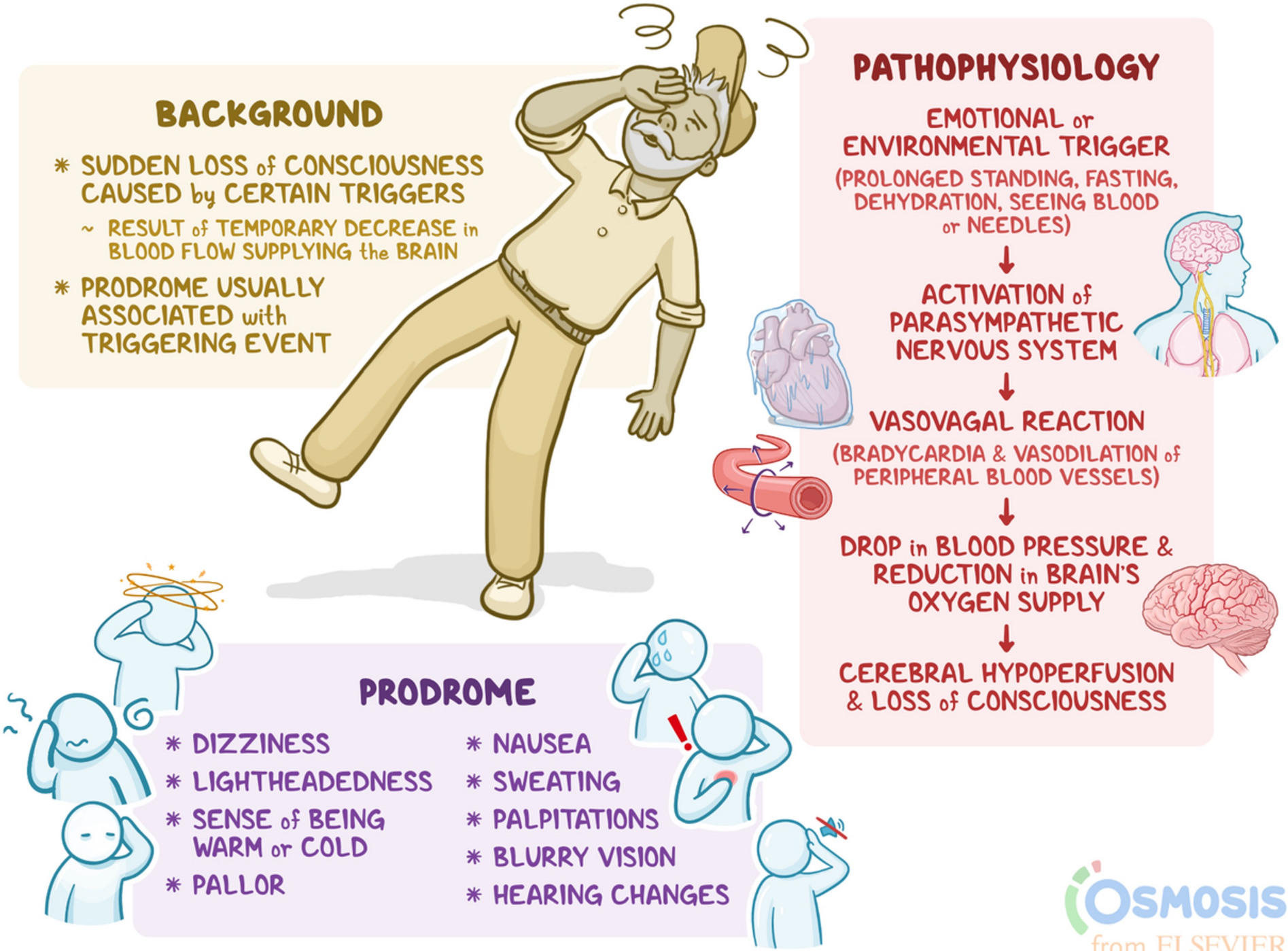


Figure 61-4. The effect of sympathectomy on blood flow in the arm and the effect of a test dose of norepinephrine before and after sympathectomy, showing *supersensitization* of the vasculature to norepinephrine.

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Vasovagal syncope



OSMOSIS from ELSEVIER



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